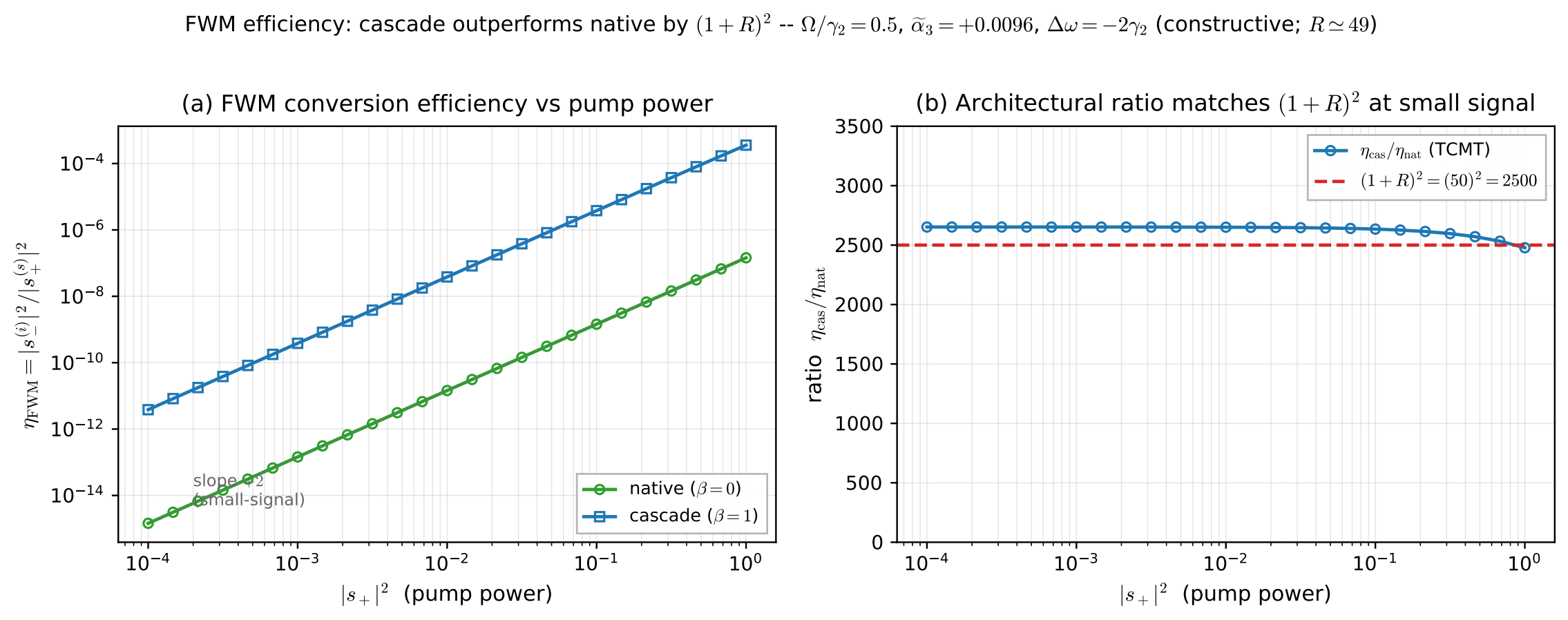
FWM efficiency: cascade outperforms native by $(1+R)^2$ -- $\Omega/\gamma_2 = 0.5$, $\widetilde{\alpha}_3 = +0.0096$, $\Delta\omega = -2\gamma_2$ (constructive; $R \simeq 49$)
(a) FWM conversion efficiency vs pump power
$\eta_{\mathrm{FWM}} = |s_-^{(i)}|^2 / |s_+^{(s)}|^2$
slope +2
(small-signal)
native ($\beta = 0$)
cascade ($\beta = 1$)
$|s_+|^2$ (pump power)
(b) Architectural ratio matches $(1+R)^2$ at small signal
ratio $\eta_{\mathrm{cas}}/\eta_{\mathrm{nat}}$
$\eta_{\mathrm{cas}}/\eta_{\mathrm{nat}}$ (TCMT)
$(1+R)^2 = (50)^2 = 2500$
$|s_+|^2$ (pump power)